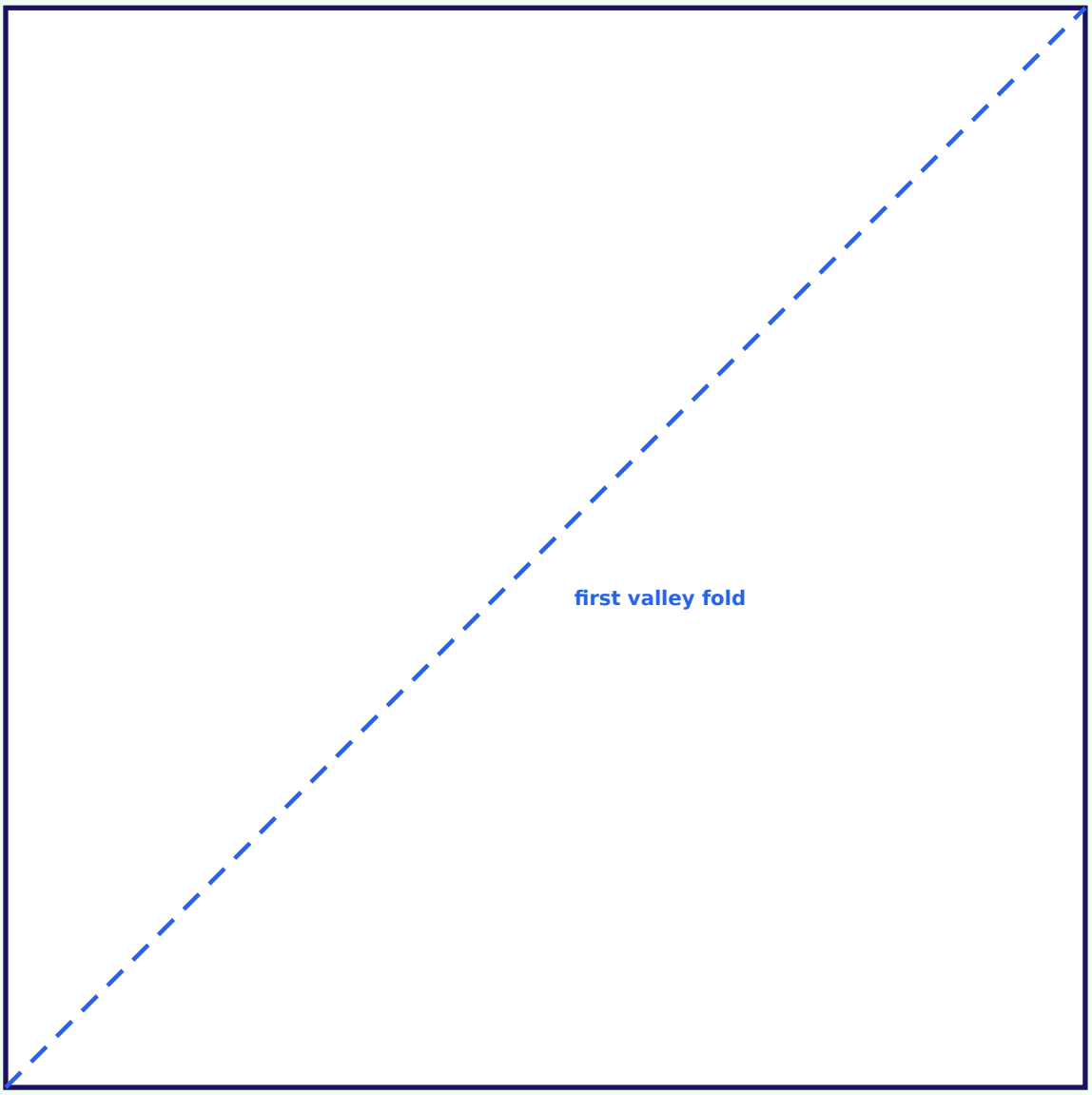
PRINT AT 100% / ACTUAL SIZE — Do NOT “fit to page”

Fold a Jumping Frog

Ages 7+ · ~15 minutes · Square 150 × 150 mm · Coding idea: stored energy + a trigger

True-size square 150 × 150 mm (cut from A4 if needed)



Turn the square like a diamond, green side down — the blue diagonal goes side to side.

150 mm

Symbol key

Valley (fold toward you)

Mountain (fold away)

↻ Turn over → Fold arrow ✓ Check the shape ⤴ Jump!

You need

- 1 square paper 150 mm (thick is best)
- Dark marker for eyes (plain paper)
- Flat, smooth table

Coding / science link

The zig-zag spring STORES energy.

Pressing the back is the TRIGGER:

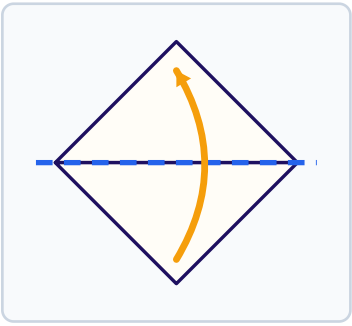
event → action. Boing!

100 mm

Measure: this bar must be exactly 100 mm. Print at 100%, not "fit to page".

20 mm

1 Big triangle

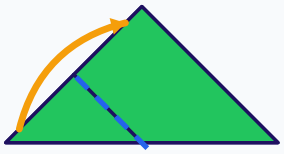


DO
Green side down, turned like a diamond. Fold the BOTTOM corner up to the top.

CHECK ✓
A big triangle, point up.

TIP
Crease with a fingernail.

2 Left corner up

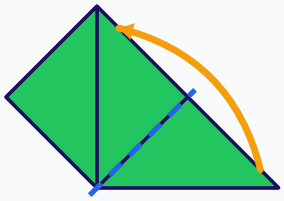


DO
Fold the left corner up to the top point.

CHECK ✓
Half a diamond.

TIP
Line the edges up first, then press.

3 Right corner up

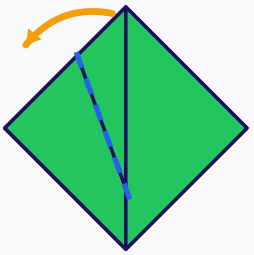


DO
Fold the right corner up to the top point too.

CHECK ✓
A diamond (a square on its point).

TIP
Same as the left, mirrored.

4 Left front leg

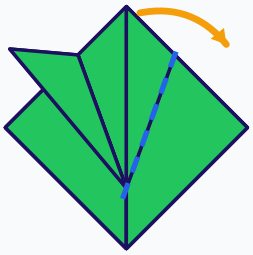


DO
Fold the tip of the left flap out to the side on the dashed line.

CHECK ✓
A leg sticks out on the left.

TIP
Fold both layers.

5 Right front leg



DO
Fold the tip of the right flap out the same way.

CHECK ✓
Two front legs!

TIP
Make both legs match.

6 Back up

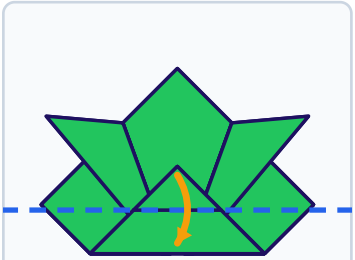


DO
Fold the bottom point up along the line.

CHECK ✓
A flat edge at the back.

TIP
Press hard — this is the spring.

7 Spring fold



DO
Fold that flap back down, halfway, on the line.

CHECK ✓
A zig-zag: the spring!

TIP
Zig-zag = stored energy.

8 Turn over



DO
Flip the frog over like a pancake.

CHECK ✓
Eyes on top, legs out the sides.

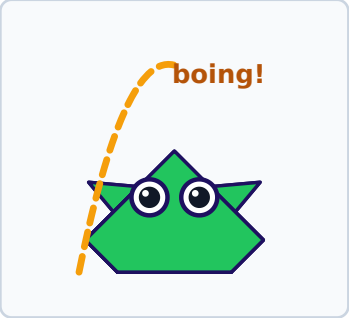
TIP
The spring is now underneath.

100 mm

Measure: this bar must be exactly 100 mm. Print at 100%, not "fit to page".

20 mm

9 Jump!

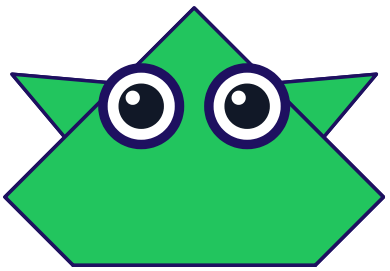


DO
Press the back edge down with one finger, then slide it off.

CHECK ✓
Boing! The frog hops.

TIP
Thicker paper jumps higher.

Finished!



How to make it jump

1. Put the frog on a smooth table, eyes up.
2. Put one finger on the back edge (the spring).
3. Push down a little so the spring squashes.
4. Slide your finger off the back — it springs up and hops!

It won’t jump?

- Crease the zig-zag harder (steps 6 and 7).
- Use thicker paper — an index card cut square is great.
- Press only the very back edge, then slide off fast.

Why does it jump?

Squashing the zig-zag STORES energy, like a coiled spring.

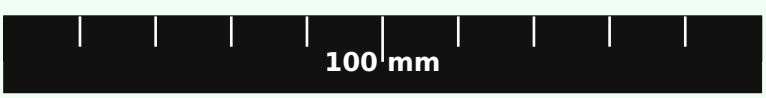
Your sliding finger is the TRIGGER — an event.

The jump is the ACTION. In code: when pressed → jump!

Frog race!

Fold two frogs and line them up. Which one jumps further?

Change ONE thing (paper, spring size) and test again — that is debugging!



Measure: this bar must be exactly 100 mm. Print at 100%, not "fit to page".

